# History-Conditioned SAC under Sensor Faults: A Pilot Study of Adaptation and Retention

**Felix Surjodinoto*, Keandre Rafael, Delroy, Samuel Philip**

Computer Science Department, BINUS Online Learning
Bina Nusantara University, Jakarta, Indonesia
ICoDiT 2026 | Paper ID: 193 | Springer Nature CCIS

**November 19, 2026**
Hybrid
(On-site & Virtual)

BINUS University @Alam Sutera,
Tangerang, Indonesia

# Motivation & Research Problem

## The Fragility of Deep Reinforcement Learning Under Unannounced Sensor Degradation

- **Physical Deployment Gap:** Deep RL policies trained in clean simulated environments suffer catastrophic failure when deployed on physical robots subject to hardware aging, calibration drift, and sensory latency.
- **Sensor Faults in the Wild:** Real-world robotic control regularly encounters zero-mean sensor noise (vibrations, thermal noise) and observation delays (transport latency, multi-rate sensor polling).
- **The Adaptation vs. Retention Dilemma:** While policies can be fine-tuned online to restore degraded return, adaptation often triggers catastrophic forgetting of nominal clean behavior.
- **Research Core Question:** Can history-conditioned sequence models (causal transformers) adapt more robustly than reactive MLPs? Does replay buffer mixing successfully retain nominal performance without causing off-policy bias?

# Problem Formulation & Fault Modeling

## Formalizing Unannounced Sensor Degradation in Continuous Locomotion (POMDP)

- **Partially Observable Formulation (POMDP):** Continuous control modeled as M = (S, A, T, R, Omega, O, gamma), where the policy observes degraded observations rather than full ground-truth states.
- **Fault Model 1: Gaussian Sensor Noise:** Observation perturbed by zero-mean Gaussian noise: s_tilde_t = s_t + e_t, e_t ~ N(0, sigma^2 I) with sigma = 0.15. High-frequency perturbation without temporal lag.
- **Fault Model 2: Observation Delay:** FIFO ring buffer of depth d = 3 (s_tilde_t = s_t-d). Introduces non-Markovian state transition dynamics and breaks immediate velocity tracking.
- **Dual Evaluation Protocol:** 1. Adaptation Return: Policy performance evaluated under active continuous sensor fault.

2. Retention Return: Retained policy competence on the original clean environment without re-training.

# Architectures & Replay Buffer Strategy

## History-Conditioned Transformer vs. Reactive MLP with Nominal Replay Mixing

- **History-Conditioned Controller (Transformer):** GPT-style causal transformer; sliding context window K = 10 steps; token embedding d_model = 64; 2 causal self-attention blocks with 4 heads; MLP projection head. 87.6k actor / 92.3k critic parameters.
- **Reactive Controller Baseline (MLP):** Standard SAC policy: 2 hidden layers with 256 units each, ReLU activations. 71.9k actor / 72.5k critic parameters. Memoryless mapping s_t -> a_t.
- **Replay Buffer Strategy (50% Mixing):** To mitigate catastrophic forgetting, 50% of the training minibatch is sampled from offline clean nominal transitions and 50% from online corrupted transitions.
- **Hypothesis Tested:** Does memory retention come at the expense of adaptation velocity, and does mixing clean and corrupted transitions distort the soft Bellman target?

# Rigorous Experimental Protocol

## 160 Policy Branches Evaluated Across HalfCheetah-v5 and Ant-v5 Benchmarks

- **Phase 1: Source Policy Training:** 8 independent source seeds per controller and task trained for 400,000 steps under nominal clean conditions (32 source baselines).
- **Phase 2: Continued Training (20k Steps):** 5 parallel branches per seed: Clean Control, Noise Only, Noise + Replay, Delay Only, Delay + Replay (80 HalfCheetah + 80 Ant = 160 branches).
- **Pre-Specified Significance Testing:** Exact Wilcoxon signed-rank tests paired on source seed; family-wise error rate controlled via Holm-Bonferroni correction (adjusted alpha floors: 0.031 and 0.0625); Cohen's d_z effect sizes and 95% bootstrap intervals.

**(a) Motivating application: TrackMania 2020**

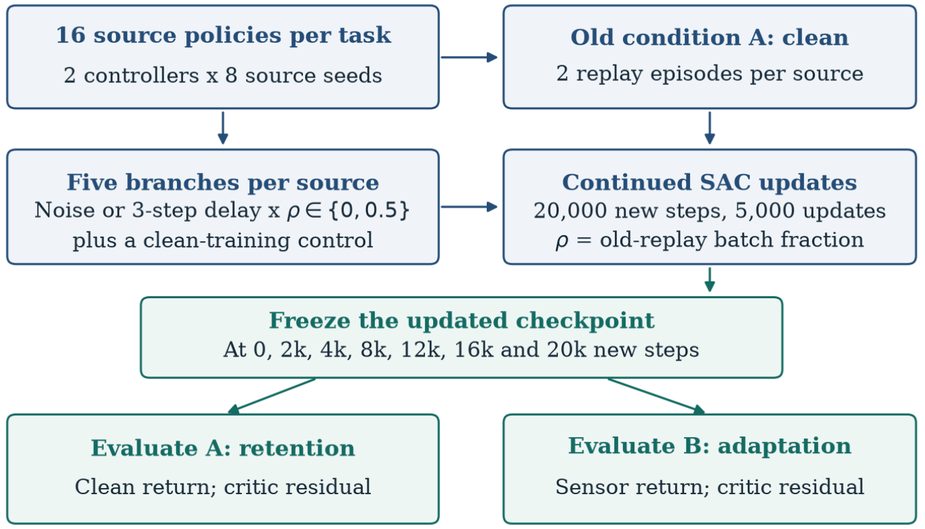



**Real-time visual control**

Goal: learn new conditions while retaining earlier control behavior.

Saved development capture

**(b) Executed study: HalfCheetah-v5 and Ant-v5**

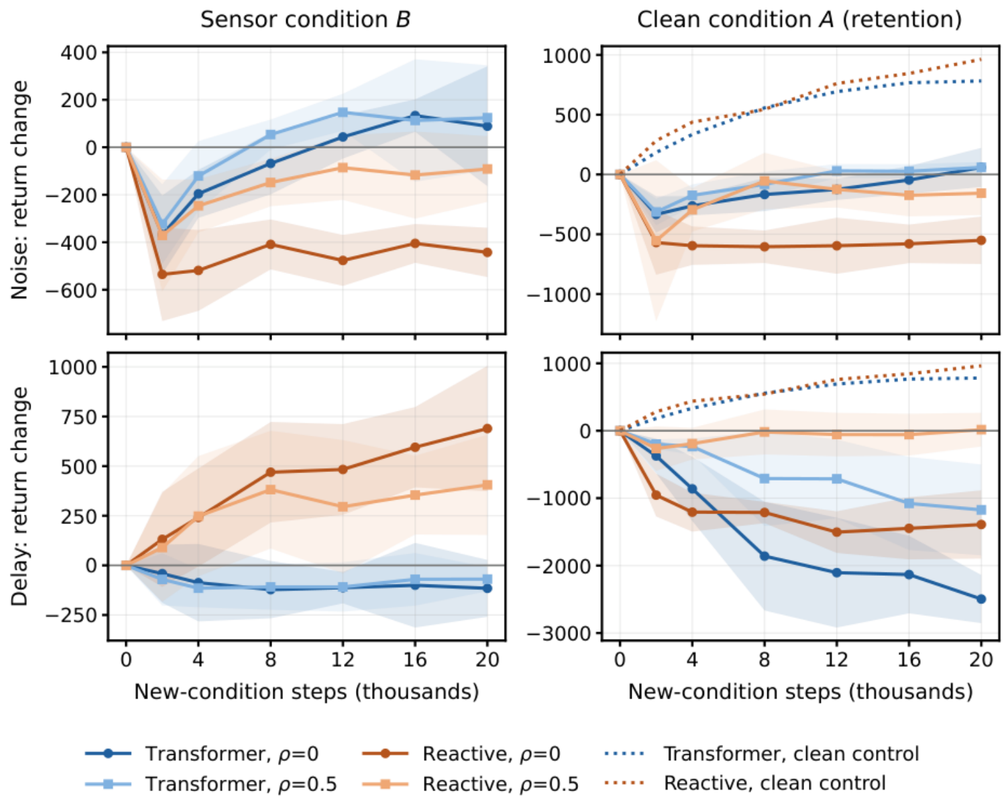



Same checkpoint for A and B; evaluation episodes stay out of replay.

# HalfCheetah-v5: Adaptation & Retention

## Divergent Dynamics Between High-Frequency Noise and Temporal Observation Delay

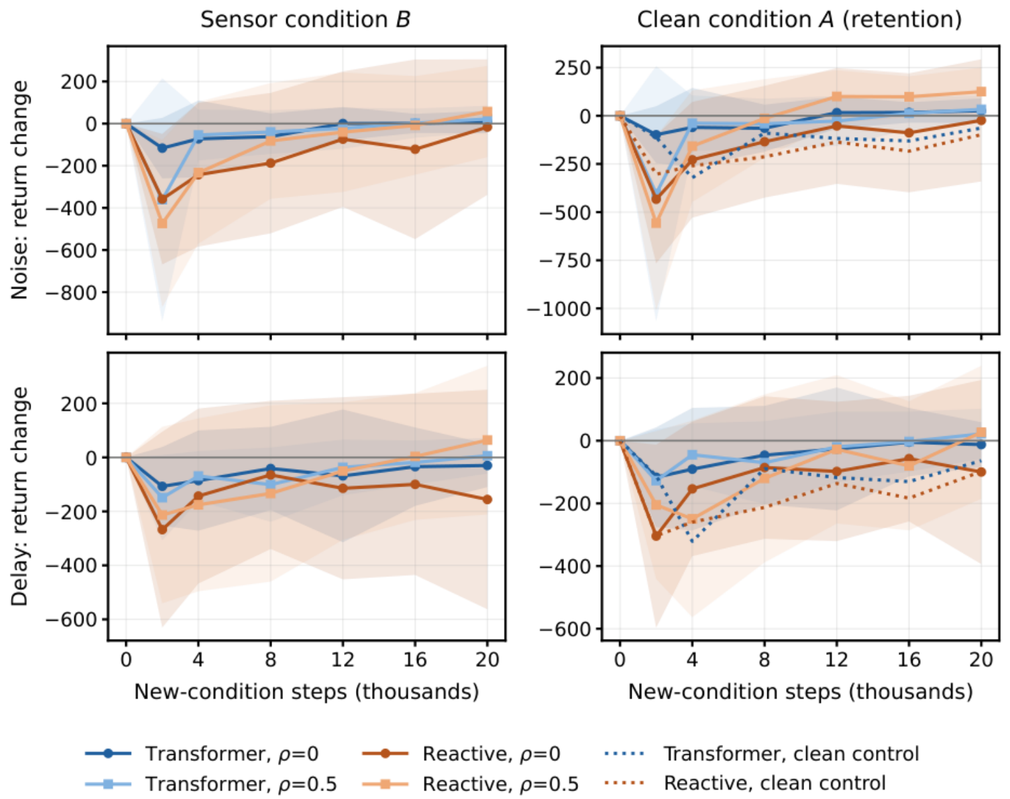
- **Sensor Noise Resilience:** Both reactive MLP and transformer recover rapid near-optimal velocity. Replay mixing successfully mitigates forgetting, preserving nominal clean returns with minimal penalty.
- **Severe Degradation Under Delay:** Observation delay inflicts massive immediate performance collapse (>2,000 return penalty). Continued training achieves only partial recovery across 20,000 steps.
- **Transformer Attention Dilemma:** Under unannounced delay, the transformer exhibits greater performance vulnerability than MLP, attending to stale historical tokens that misrepresent current plant physics.



# Ant-v5: Morphology & Task Sensitivity

## Morphological Postural Convergence and Re-Optimization Transients

- **Morphological Standing Posture:** Under the standardized 400,000-step source recipe, all 8 Ant seeds converged to an ultra-stable standing posture (forward velocity within +/-0.007 m/s).
- **Fault Sensitivity Finding:** Because standing requires negligible dynamic velocity feedback, sensor noise and delay inflicted negligible performance penalties. Reported transparently rather than retroactively tuning hyperparameters.
- **Warm-Start Re-Optimization Dip:** All Ant branches (including clean controls) experienced an initial transient return drop in the first 2,000 steps, proving that optimizer re-initialization rather than sensor degradation caused the dip.



# Mechanistic Critic Diagnostic

## Soft Bellman Target Distortion Explains Replay Failure Under Observation Delay

- **Held-Out Critic Evaluation Framework:** Both soft Q-critics evaluated on held-out clean and corrupted trajectories at source checkpoint and after 20,000 steps. Evaluated soft Bellman residual and Monte-Carlo value-bias gap (Bias_corrupt - Bias_clean).
- **Massive Source Value Overestimation:** Source critics severely overvalued delayed history states (+171 and +268 return units in HalfCheetah).
- **Distortion From Replay Mixing:** Training on corrupted data alone shrank the bias gap to +27 and +114. Mixing 50% clean replay maintained the gap elevated at +89 and +204 (paired effect: +62 [16, 109] and +90 [41, 139], 7/8 seeds positive).
- **Core Theoretical Insight:** Mixing stale clean transitions creates an off-policy distribution mismatch that actively distorts the soft Bellman target, proving why replay harmed adaptation under delay.

# Real-Time Edge Viability: 500 Hz Benchmark

## Sub-Millisecond Inference Proves Feasibility for High-Speed Robotic Control

| Platform | Controller | Threads | Mean Latency | 99th %ile | Max Latency | 500 Hz Viable? |
|---|---|---|---|---|---|---|
| CPU (i7-8700) | Transformer Actor | 1 thread | 0.58 ms | 0.66 ms | 1.56 ms | YES (Budget < 2.0 ms) |
| CPU (i7-8700) | Reactive MLP Actor | 1 thread | 0.09 ms | 0.12 ms | 0.28 ms | YES |
| CPU (i7-8700) | Transformer Actor | 4 threads | 0.64 ms | 0.78 ms | 1.82 ms | YES |
| GPU (RTX 3070 Ti) | Transformer Actor | CUDA | 1.64 ms | 1.92 ms | 3.14 ms | MARGINAL (Kernel overhead) |

- **10,000 Timed Forward Passes:** Measured at deployment batch size 1 with real model inputs (excluding physics engine simulation overhead).
- **Strict Real-Time Compliance:** Transformer inference on single-threaded CPU consistently completes in under 1.56 ms, comfortably within the 2.0 ms cycle budget of 500 Hz robotics.
- **Edge Architecture Takeaway:** Single-threaded CPU execution outperforms GPU at batch size 1, confirming suitability for embedded robot microcontrollers without dedicated discrete accelerators.

# Conclusions & Key Insights

## Actionable Takeaways for Fault-Tolerant Reinforcement Learning in Physical Systems

- **1. History Conditioning is Fault-Specific:** Causal transformers act as effective temporal low-pass filters under high-frequency noise, but suffer severe attention confusion under observation delay unless delay is explicitly represented.
- **2. Replay Buffer Trade-Off:** 50% nominal replay mixing successfully prevents catastrophic forgetting under noise, but induces severe soft Bellman target distortion and off-policy bias under non-Markovian delay.
- **3. Strict Real-Time Deployment:** Transformer history conditioning requires only 87k parameters and executes in 0.58 ms on a single CPU thread, satisfying real-time 500 Hz robotics.
- **4. Open Science & Provenance:** All code, environment wrappers, seed protocols, checkpoints, and evaluation records are publicly released at https://github.com/Fane-Nathan/tmrl.

# Thank You & Discussion

## History-Conditioned SAC under Sensor Faults: A Pilot Study of Adaptation and Retention


- **Authors:** Felix Surjodinoto*, Keandre Rafael, Delroy, Samuel Philip
- **Affiliation:** Computer Science Department, BINUS Online Learning, Bina Nusantara University, Jakarta, Indonesia
- **Primary Contact:** felix.surjodinoto@binus.ac.id

- **Artifact Repository:** https://github.com/Fane-Nathan/tmrl (Code, Checkpoints, Evaluation Logs)

- **Conference:** 2nd International Conference on Digital Technology (ICoDiT 2026) | Paper ID: 193